CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

[redacted]

Lecture No.- 09

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Recap of Previous Lecture

- **Topic** Definition in Connectivity
- **Topic** Connected vs Disconnected
- **Topic** Range of Edges
- **Topic** Concepts of Tree
- **Topic** Connectivity Theorem

Topics to be Covered

- **Topic** Analysis in Connectivity
- **Topic** Various Definition in Connectivity
- **Topic** Edge Connectivity
- **Topic** Vertex Connectivity
- **Topic** Largest Inequality Theorem

Topic : Connectivity in Graphs

Range of edges $(k = 1)$

$1 \equiv 2 \equiv 3 \equiv 4$

Tree.

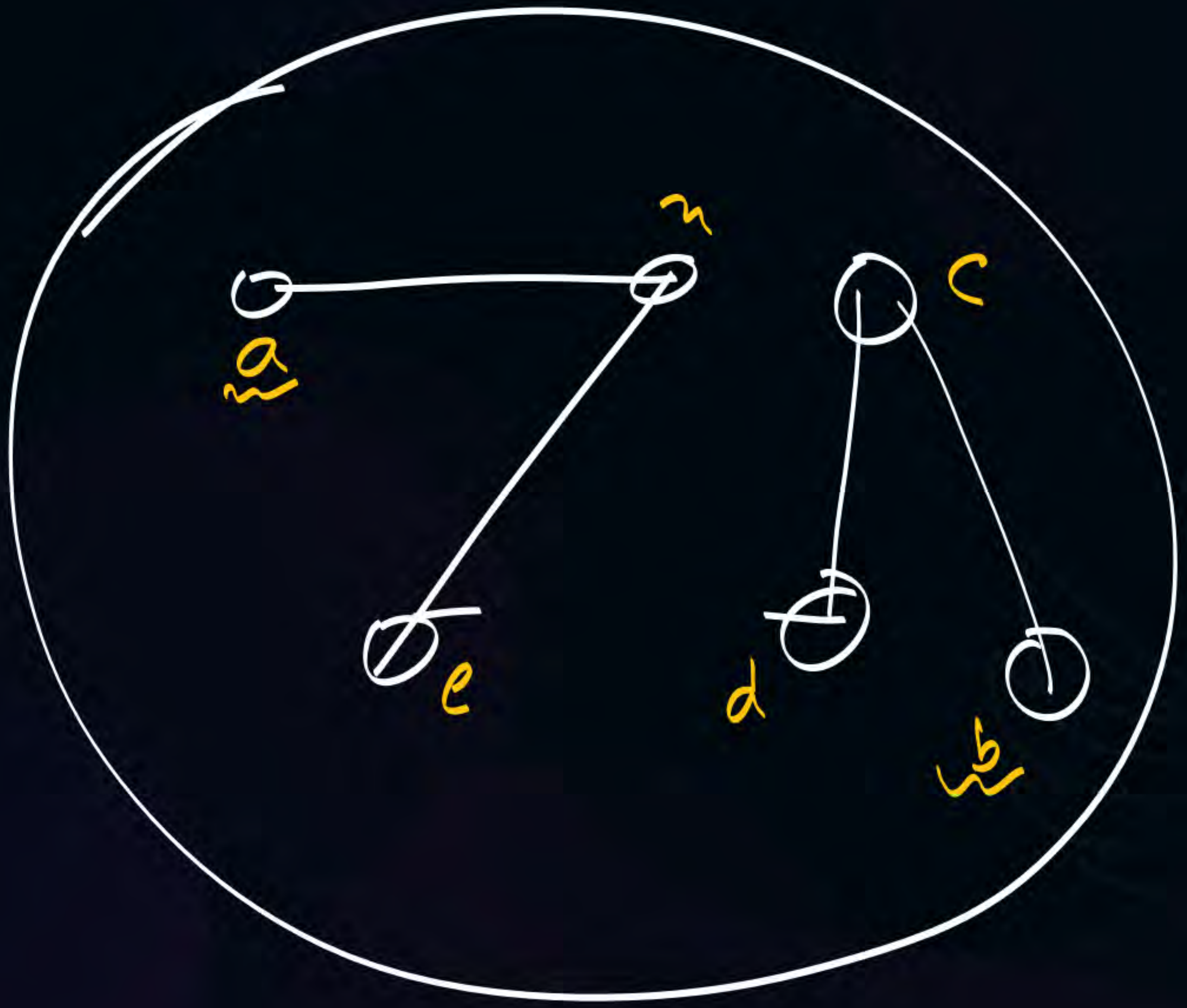
1. $n - 1$ + connected. $e \le \frac{n(n-1)}{2}$ (complete Graph)

Acyclic Graph.

2. G is connected & $\boxed{\text{doesnot contains cycle.}}$

3. unique Path bet^n all pair of vertices.

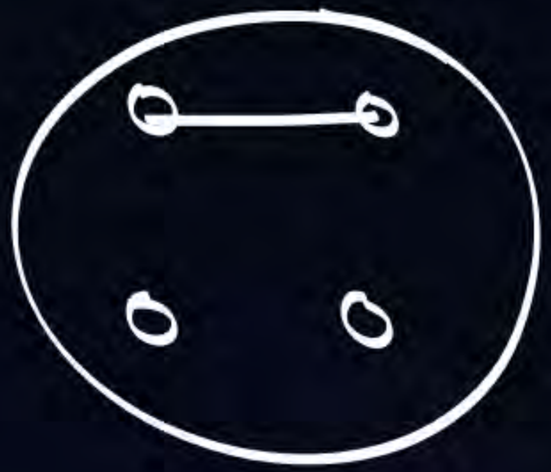
4. minimally connected.



a
x
c
e
d
b

X $n = 4$ $e = 0$

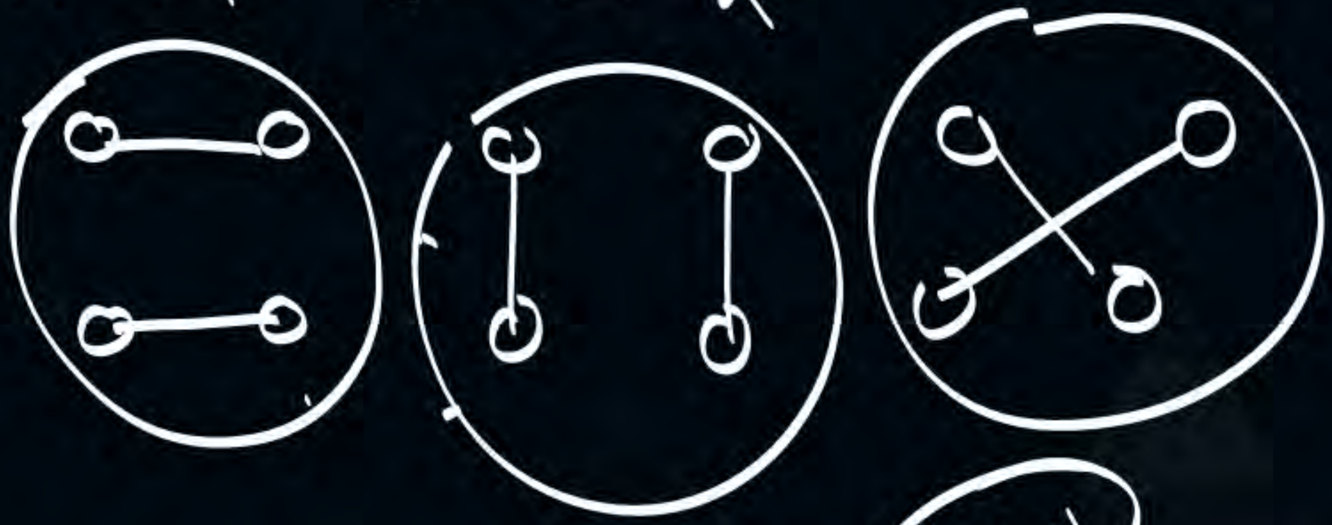
$n = 4.$ $e = 1$



X

min no. of edges + connected.

$n = 4$ $e = 2$ X



$n = 4$ $e = 3.$

$e = n-1$ → min no. of edges + connected.

Range of edges $(k \geq 2)$ (Disconnected Graph)

$$n-k \leq e \leq \frac{(n-k)(n-k+1)}{2}$$

Connected. min no. of edges in G with n vertices. **Disconnected Graph.**

Connected + min = Tree

Connected min.

C + min.

C + min.

$K = 4$.

$e_1 = n_1 - 1$	$n_2 - 1$	$n_3 - 1$	$n_4 - 1$
Total vertices in 1st component $= n_1$	n_2	n_3	n_4

Total vertices $= n$.

$n_1 + n_2 + n_3 + n_4 = n$.

min no. of edges = collection of Trees = Forest

$$e = n_1 - 1 + n_2 - 1 + n_3 - 1 + n_4 - 1$$

$$= (n_1 + n_2 + n_3 + n_4) - 4$$

$$= n - 4 = n - K$$

max no. of edges $= \sum_{i=1}^{K} \frac{n_i(n_i - 1)}{2}$.

What will be min no. of edges & max no. of edges in Graph.

A) $n = 7$ $K = 2$.

B) $n = 11$ $K = 3$

C) $n = 10$ $K = 1$.

	min	max.
a)	$e = 5$	$e = 15$
b)	$e = 8$	$e = 36$.
c)	$e = 9$.	$e = 45$

$$e = \frac{(n-K)(n-K+1)}{2}. \qquad K = 1.$$

$$= \frac{(n-1)(n-1+1)}{2} = \frac{(n-1)(n)}{2}.$$

2 mins Summary

Topic One

Topic Two

Topic Three

Topic Four

Topic Five

THANK - YOU